

Austo Automobile Dealership

Market Analysis & Strategic Business Report

PGP-DGGA'26 Project 1: Exploratory Data Analysis

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1. Executive Summary

This report provides a comprehensive data-driven analysis of the Austo Automobile dealership's customer base and sales performance. By interrogating a dataset of 1,581 transactions, we have identified core market segments, evaluated the impact of finance and lifecycle stage of customer on revenue, and tested existing market assumptions.

Key Finding: The analysis debunked the prevalent 'Sheldon Cooper' theory that salaried males are prime targets for SUVs. In reality, this segment shows an overwhelming preference for Sedans, which represent the dealership's most consistent revenue driver across nearly all demographics.

2. Data Overview

A thorough diagnosis of the 1,581-record dataset was performed to ensure statistical validity. Major treatment steps included:

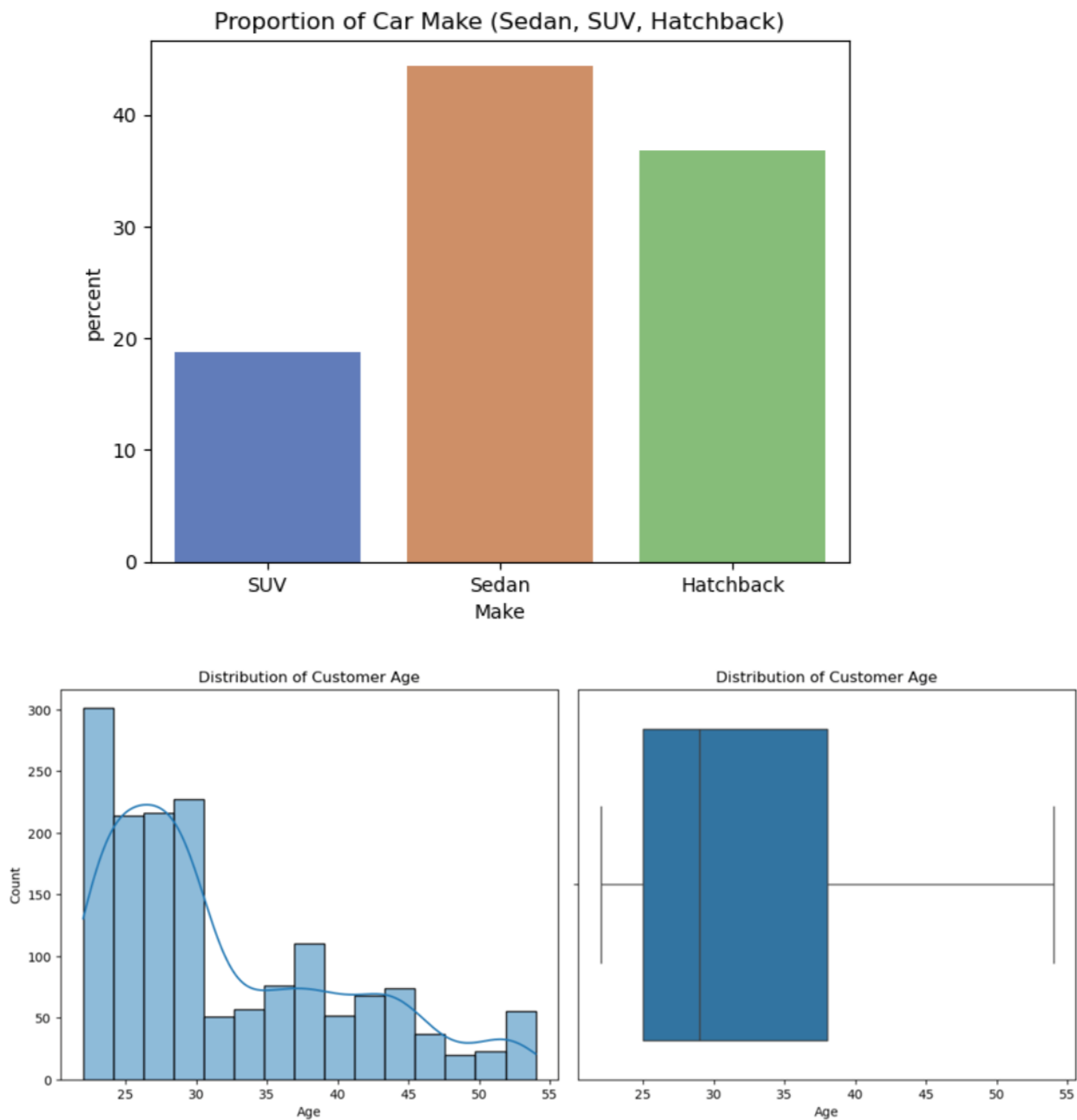
- **Gender Data Treatment:** Unified inconsistent spellings ('Femal', 'Femle') and categorized missing values as 'Unknown'.
- **Outlier Management:** Identified a 'Right Skew' in car prices (extending to \$70,000). These were preserved as they represent the high-value luxury segment.

Metric	Value
Total Transactions	1,581
Median Car Price	\$31,000
Primary Profession	Salaried (57%)

3. Customer Profile (Univariate Analysis)

The core customer base consists of young to mid-career professionals, primarily aged between 25 and 38. The gender distribution is heavily skewed toward Male buyers (~75%).

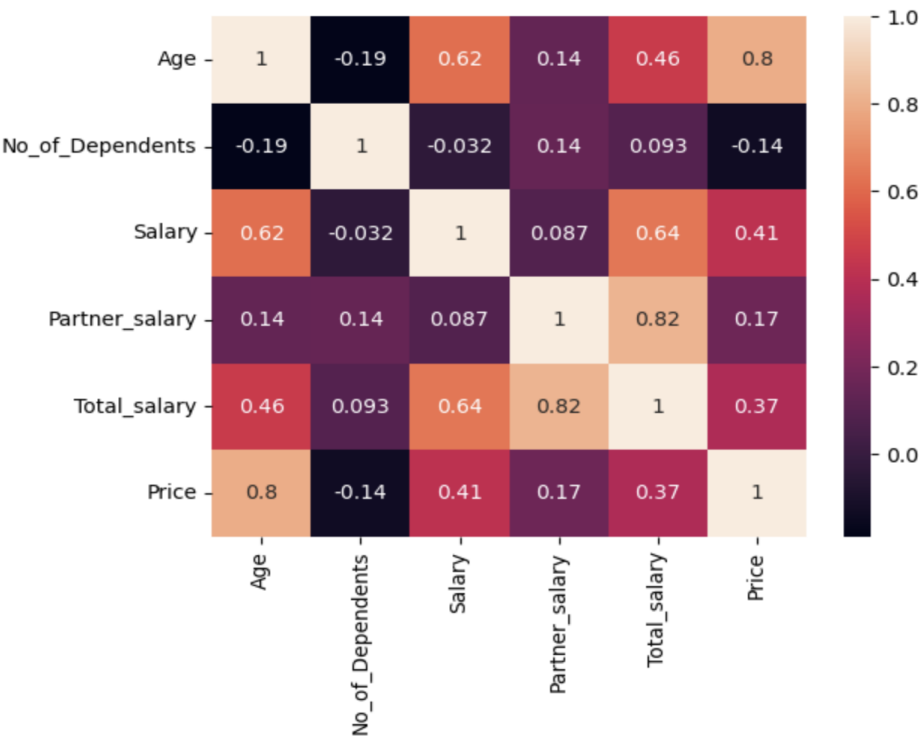
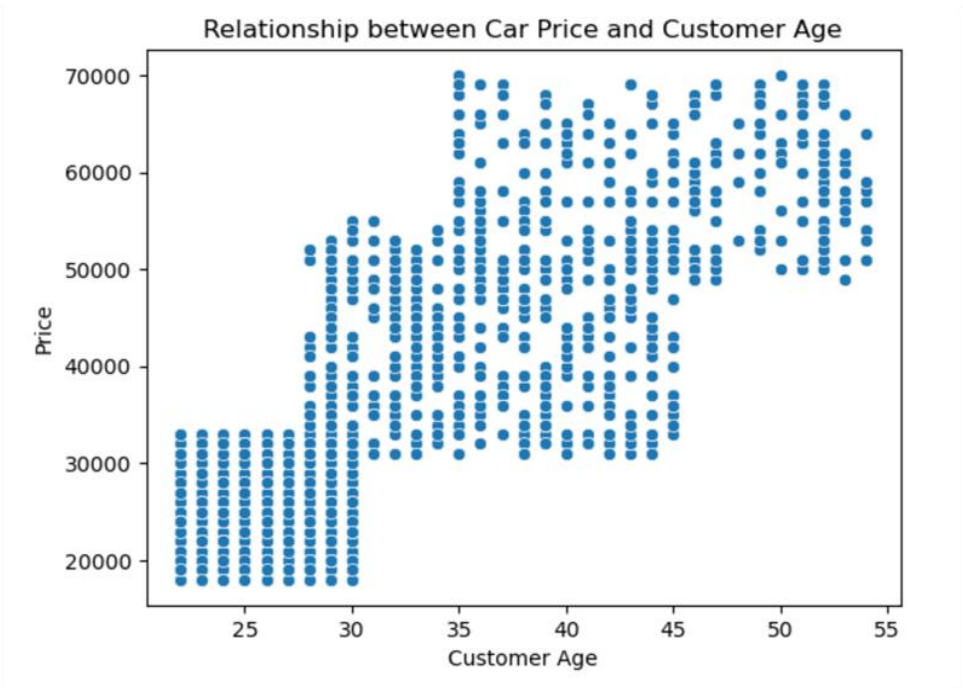
Market Preference: Sedans (45%), Hatchbacks (35%), and SUVs (20%).



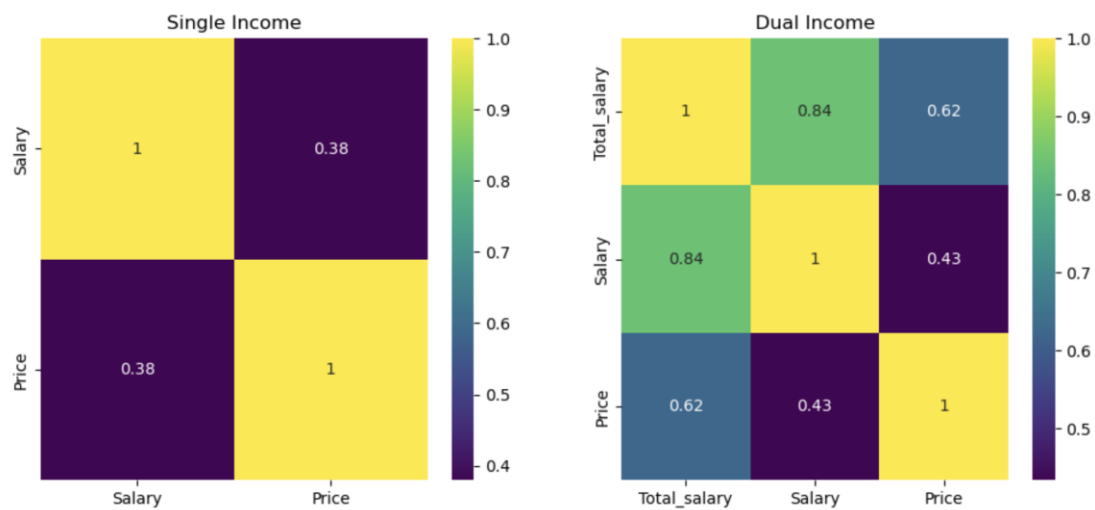
4. Key Business Drivers (Bivariate Analysis)

Strongest Predictors of Spend:

Age Factor (0.80 Correlation): As customers mature, they upgrade to higher priced cars.



Income: In dual-income households, total household income shows a stronger relationship with car price (~0.62) compared to individual salary in single-income households (~0.38), indicating that combined earning capacity better explains purchasing power in these segments.



5. Critical Business Questions

1. Do men tend to prefer SUVs more compared to women?

Answer: *No, Men do not tend to prefer SUVs more compared to women.*

In this dataset, male customers are more inclined toward Hatchbacks and Sedans, while female customers show a significantly higher preference for SUVs.

Specifically, ~52% of female customers purchase SUVs compared to only ~10% of male customers, indicating that women are substantially more likely to opt for SUVs than men.

	Make	Hatchback	SUV	Sedan
Gender				
Female		0.045593	0.525836	0.428571
Male		0.471226	0.098415	0.430359
Unknown		0.037736	0.113208	0.849057

2. What is the likelihood of a salaried person buying a Sedan?

Answer: The likelihood of a salaried person buying a Sedan is approximately 44%.

This indicates that Sedans are the most preferred car type among Salaried customers, slightly higher than Hatchbacks (~ 32 %) and significantly higher than SUVs (~ 23 %)

This indicates that salaried customers prefer the mid-price segment, which offers a balance between affordability and value, rather than opting for the premium-price segment.

	Make	Hatchback	SUV	Sedan
Profession				
Business		0.423358	0.129927	0.446715
Salaried		0.325893	0.232143	0.441964

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

Answer: The data does not support Sheldon Cooper's claim.

Among salaried male customers, only ~13% purchase SUVs, while ~44% opt for Sedans. This indicates that salaried males are significantly more inclined towards Sedans rather than SUVs, making them a weaker target segment for SUV sales.

In essence, the claim is contradicted by data.

Profession	Make	Hatchback	SUV	Sedan
	Gender			
Business	Female	0.000000	0.523810	0.476190
	Male	0.516995	0.059034	0.423971
	Unknown	0.047619	0.047619	0.904762
Salaried	Female	0.066964	0.526786	0.406250
	Male	0.431250	0.132812	0.435937
	Unknown	0.031250	0.156250	0.812500

4. How does the the amount spent on purchasing automobiles vary by gender?

Answer: A typical female customer spent substantially higher on car compared to a typical Male customer.

The median price spent by a Female customer is 49k USD while a Male typically spent 29k USD indicating a noticeable difference in spending patterns. However, the Female customer sample size is significantly small compared to Male customers, so this observation should be interpreted with caution.

	count	mean	std	min	25%	50%	75%	max
Gender								
Female	329.0	47705.167173	11244.836378	20000.0	38000.0	49000.0	55000.0	69000.0
Male	1199.0	32817.347790	12299.239195	18000.0	24000.0	29000.0	37500.0	70000.0
Unknown	53.0	23339.622642	10308.591257	18000.0	18000.0	18000.0	19000.0	50000.0

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Answer: Individuals with a personal loan spent a total of 27.3M USD, accounting for 48.5% of the dealership's total revenue.

On an individual basis, they spent an average of ~\$31k, which is consistent with cash buyers. This indicates that personal loan status does not significantly influence car purchase spending. Instead, purchasing behavior appears to be driven more by overall affordability than by loan dependency.

Amount Spent by individuals with Personal loan: 27290000

Total revenue of Dealership: 56280000

	count	mean	std	min	25%	50%	75%	max
Personal_loan								
No	789.0	36742.712294	14534.344526	18000.0	25000.0	32000.0	49000.0	70000.0
Yes	792.0	34457.070707	12578.780338	18000.0	24000.0	31000.0	45000.0	70000.0

6. How does having a working partner influence the purchase of higher-priced cars?

Answer: Having a working partner does not significantly influence the purchase of higher-priced cars.

The median price for both single-income and dual-income households remains identical at ~31k USD, with mean values also showing minimal variation.

While dual-income households show a slightly higher preference for Sedans, there is no meaningful increase in SUV purchases, indicating that additional income does not directly translate into higher-priced car purchases in this dataset.

Partner working status influences what type of car customers lean toward, but does not significantly impact how much they spend.

	Make	Hatchback	SUV	Sedan
Partner_working				
No		0.394109	0.201964	0.403927
Yes		0.346774	0.176267	0.476959

	count	mean	std	min	25%	50%	75%	max
Partner_working								
No	713.0	36000.000000	13817.734086	18000.0	25000.0	31000.0	48000.0	70000.0
Yes	868.0	35267.281106	13479.532555	18000.0	24000.0	31000.0	46000.0	70000.0

6. Actionable Insights & Strategic Recommendations

1. Age is the strongest driver of car purchase behavior Customers tend to move toward higher-priced vehicles as they age, indicating a shift in preferences and lifestyle.
 - **Recommendation:** Focus on customers aged 40+ for mid-to-premium segments, positioning vehicles around comfort, status, and long-term value rather than just price. Target messaging by life stage, not just income:
 - Younger → affordability, practicality
 - Mid-age → upgrade, family needs
 - Older → comfort, status
2. Gender differences exist but should be interpreted cautiously Female customers show a higher median spend compared to male customers. However, this insight is influenced by sample imbalance and should not be overgeneralized.
 - **Recommendation:** Position mid-segment cars as “value for money” to appeal to the dominant male segment, while maintaining aspirational positioning for higher segments without over-targeting based on gender alone.
3. Loan status does not significantly impact spending behavior Customers with and without personal loans exhibit similar spending patterns, indicating that financing does not strongly influence the price segment chosen.
 - **Recommendation:** Offer flexible financing and EMI options across all segments rather than targeting specific loan-based customer groups.
4. Sedans are consistently popular among all income, professions, and demographics, making them widely appealing. Sedans attract both ambitious buyers and those wanting good value.
 - **Recommendation:** Position Sedans as “reliable premium” or “executive comfort” to capture both aspirational and practical buyers.

5. Dual-income households do not significantly increase spending but show a tilt toward Sedans While dual-income does not lead to higher spending, there is a slight preference shift toward Sedans.
 - **Recommendation:** Target dual-income households with messaging around practicality, comfort, and everyday usability rather than premium positioning alone.
6. Income drives upgrade path across segments Customers transition from Hatchbacks to Sedans (and slightly toward SUVs) as income increases.
 - **Recommendation:** Design tiered marketing:
 - Low income → “affordable, practical” (Hatchback)
 - Mid income → “upgrade, comfort” (Sedan)
 - High income → “status + performance” (SUV/Sedan premium)

7. Conclusion

Customer purchasing behavior at Austo Automobile is primarily driven by age and income, which together determine both the price and segment of cars purchased. As customers progress through life stages and increase their earning capacity, they move away from entry-level hatchbacks toward mid-to-high segment vehicles.

Sedans emerge as the most consistently preferred car type across customer groups, acting as a stable and central segment in the market. While demographic and financial variables such as gender, profession, and loan status introduce some variation, their overall impact on purchasing decisions is limited.

Overall, car purchase behavior is shaped more by lifecycle progression and affordability than by isolated customer attributes, providing clear direction for targeted marketing and product positioning strategies.

Thank you!